

RAMCO INSTITUTE OF TECHNOLOGY

**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING**

COURSE OUTCOMES

REGULATION 2013

Course Code & Title : HS6151 Technical English-I Year / Semester : I Year/I Semester Course Index : C101	
C101.1	Explain clearly, confidently, comprehensibly, and communicate with one or many listeners using appropriate communicative strategies.
C101.2	Construct cohesively and coherently and flawlessly avoiding grammatical errors, using a wide vocabulary range, organizing their ideas logically on a topic.
C101.3	Organize different genres of texts adopting various reading strategies.
C101.4	Distinguish and comprehend different spoken discourses/excerpts in different accents.
C101.5	Improve listening to different accents, speeches and presentations.
Course Code & Title : MA6151 Mathematics - I Year/ Semester : I Year/I Semester Course Index : C102	
C102.1	Infer canonical form from quadratic form using the concepts of Eigen values and Eigen vectors.
C102.2	Test the convergence of sequences and series.
C102.3	Apply the techniques of differential calculus to find the evolute and envelope of curves.
C102.4	Determine the maxima and minima of functions of two variables using partial derivatives.
C102.5	Find the area enclosed by plane curves and volume of solids using multiple integrals.
Course Code & Title : PH6151 Engineering Physics - I Year/ Semester : I Year/I Semester Course Index : C103	
C103.1	Describe the crystal structures and various crystal growth techniques.
C103.2	Analyze the elastic nature of materials and thermal behaviour of materials.
C103.3	Apply the knowledge of quantum mechanics and classical mechanics in addressing the problems related to science and technology.
C103.4	Apply the knowledge about designing an auditorium with good acoustical properties and make use of Ultrasonics and its applications in various fields.
C103.5	Illustrate the advantages of optical communication using LASER.
Course Code & Title : CY 6151 Engineering Chemistry -I Year/ Semester : I Year/I Semester Course Index : C104	
C104.1	Describe the methods of polymerization, types, properties and uses of polymers.
C104.2	Illustrate the concepts of basic thermodynamics and problem solving skills in various disciplines of Engineering.
C104.3	Discuss the laws of photochemistry in recognizing the interaction of light with matter and its applications in luminescence and spectroscopy.
C104.4	Review the use of phase rule in identifying its applications in metallurgy and alloys.
C104.5	Summarize the basic knowledge in Nano-chemistry and distinguish the existing technology with nanotechnology.

Course Code & Title : GE6151 Computer Programming	
Year/ Semester : I Year/I Semester	
Course Index : C105	
C105.1	Elaborate the organization of a digital computer and design the solution for simple computing problems using algorithm, flowchart and pseudo code
C105.2	Apply different looping structures to solve simple scientific and statistical problems
C105.3	Devise solutions for simple problems using array and strings
C105.4	Demonstrate the usage of Dynamic memory allocation and pointer variables
C105.5	Illustrate the concepts of structures and unions with example programs
Course Code & Title : GE6152 Engineering Graphics	
Year/ Semester : I Year/I Semester	
Course Index : C106	
C106.1	Sketch the conic sections, special curves, and draw orthographic views from pictorial views and models.
C106.2	Apply the principles of orthographic projections of points in all quadrants, lines and planes in first quadrant.
C106.3	Sketch the projections of simple solids like prisms, pyramids, cylinder and cone and obtain the traces of plane figures.
C106.4	Practice the sectional views of solids like cube, prisms, pyramids, cylinders & cones and extend its lateral surfaces.
C106.5	Sketch the perspective projection of simple solids, truncated prisms, pyramids, cone and cylinders and sketch the isometric projection of simple machine parts.
Course Code & Title : GE6161 Computer Programming Laboratory	
Year/ Semester : I Year/I Semester	
Course Index : C107	
C107.1	Understand the usage of office automation tools.
C107.2	Apply good programming design methods for program development.
C107.3	Design and implement the C programs for simple applications.
C107.4	Develop and implement the recursive programs.
C107.5	Implement the c programs with the help of structures and unions.
Course Code & Title : GE6162 Engineering Practices Laboratory	
Year/ Semester : I Year/I Semester	
Course Index : C108	
C108.1	Construct carpentry components and pipe connections including plumbing works.
C108.2	Use welding equipment's to join the structures
C108.3	Illustrate the basic machining operations
C108.4	Construct the models using sheet metal works
C108.5	Describe centrifugal pump, Air conditioner, operations of smithy, foundry and fittings.
C108.6	Construct the basic Electrical and Electronics circuits.
C108.7	Examine the different types of electronic circuits and components.
C108.8	Explain the electrical safety rules, grounding, general house wiring.
C108.9	Perform soldering in various electronic circuits.
C108.10	Illustrate the basic operation of domestic electrical appliances.
Course Code & Title : GE6163 Physics and Chemistry Laboratory - I	

Year/ Semester : I Year/I Semester	
Course Index : C109	
C109.1	The hands on exercises undergone by the students will help them to apply physics principles of optics and thermal physics to evaluate engineering properties of materials.
C109.2	Perform the quantitative chemical analysis of chloride and dissolved oxygen.
C109.3	Determine the amount of acids by using the instruments of conductivity meter and pH meter.
Course Code & Title : HS 6251 Technical English-II	
Year/ Semester : I Year/ II Semester	
Course Index : C110	
C110.1	Speak convincingly, express their opinions clearly, initiate a discussion, negotiate, argue using appropriate communicative strategies.
C110.2	Write effectively and persuasively and produce different types of writing such as narration, description, exposition and argument as well as creative, critical, analytical and evaluative writing.
C110.3	Read different genres of texts, infer implied meanings and critically analyse and evaluate them for ideas as well as for method of presentation.
C110.4	Listen/view and comprehend different spoken excerpts critically and infer unspoken and implied meanings.
C110.5	Students will become accomplished, active readers and will be able to write effectively for a variety of professional and social settings
Course Code & Title : MA6251 Mathematics - II	
Year/ Semester : I Year/ II Semester	
Course Index : C111	
C111.1	Apply the knowledge of vector calculus in engineering disciplines.
C111.2	Solve ordinary differential equations that model the engineering problems.
C111.3	Find the Laplace transform of functions and solve the ordinary differential equations using Laplace transform.
C111.4	Construct analytic functions and apply the knowledge of conformal mappings in engineering disciplines.
C111.5	Evaluate contour integration and apply it in engineering problems.
Course Code & Title : PH6251 Engineering Physics - II	
Year/ Semester : I Year/ II Semester	
Course Index : C112	
C112.1	Describe the conducting materials and their properties.
C112.2	Analyze the semiconductors and able to differentiate various types of semiconductors.
C112.3	Apply the knowledge of magnetic and superconducting materials for modern day to day applications.
C112.4	Explain the properties and applications of dielectrics.
C112.5	Apply the knowledge about the modern engineering materials for various applications.
Course Code & Title : CY6251 Engineering Chemistry - II	
Year/ Semester : I Year/ II Semester	
Course Index : C113	
C113.1	Comprehend the importance of water technology in the purification of water and

	its domestic and industrial applications.
C113.2	Explain the principles of electrochemistry, the factors affecting corrosion and The prevention of corrosion.
C113.3	Understand the fundamentals of different alternative sources of energy and the generation processes.
C113.4	Acquire the knowledge on different types of engineering materials and their applications.
C113.5	Learn the know-how of industrial techniques of petroleum processing and the determination of calorific values and combustion parameters.
Course Code & Title : GE6251 Basic Civil and Mechanical Engineering Year/ Semester : I Year/ II Semester Course Index : C114	
C114.1	To explain the usage of construction material and proper selection of construction materials.
C114.2	To design building structures.
C114.3	Identify the components used in power plants
C114.4	Describe the working principles of Internal Combustion Engines
C114.5	Explain the principle of Refrigeration and Air Conditioning System
Course Code & Title : EE6201 Circuit Theory Year/ Semester : I Year/ II Semester Course Index : C115	
C115.1	Analyze circuits with ideal, independent, and controlled voltage and current sources.
C115.2	Apply ohm's law, Kirchhoff's laws, Network theorems to the analysis of AC and DC circuits.
C115.3	Solve first and second order AC and DC Circuits for Steady – State and transient response in the time domain and frequency domain.
C115.4	Evaluate the response of the coupled circuits and two port networks
C115.5	Estimate the power consumed by the three phase balanced and unbalanced circuits.
Course Code & Title : GE6262 Physics and Chemistry Laboratory - II Year/ Semester : I Year/ II Semester Course Index : C116	
C116.1	Test materials by using their knowledge of applied physics principles in optics and properties of matter.
C116.2	Determine the hardness, alkalinity and metal ion content in the water samples by volumetric titration.
C116.3	Estimate the water quality parameters by potentiometer, conductometer and flame photometer.
Course Code & Title : GE6263 Computer Programming Laboratory Year/ Semester : I Year/ II Semester Course Index : C117	
C117.1	Know the Structure of Unix Operating System and practice basic commands in Unix Shell
C117.2	Devise solutions for simple problems using conditional statements and loops in Unix shell
C117.3	Illustrate the concepts of Dynamic Memory Allocation, Pointers and File

	Allocation in C programming under Unix
Course Code & Title : EE6211 Electric Circuits Laboratory Year/ Semester : I Year/ II Semester Course Index : C118	
C118.1	Solve the electrical circuit problems using circuit theorems and laws
C118.2	Verify the various circuit theorems experimentally as well as in simulation
C118.3	Determine the frequency response and transients in passive circuits
C118.4	Simulate the resonance circuits for several applications such as designing of tuning circuit, signal processing and voltage magnification
C118.5	Estimate the solution of three phase circuits under both balanced and unbalanced conditions
Course Code & Title : MA6351 Transforms and Partial Differential Equations Year/ Semester : II Year/ III Semester Course Index : C201	
C201.1	Solve the Partial Differential Equations.
C201.2	Determine the Fourier series expansion of functions and hence evaluate the value of infinite series.
C201.3	Apply the method of separation of variables to solve one dimensional wave equation, one dimensional heat equation and two dimensional heat equation.
C201.4	Find the Fourier transform of functions and also evaluate definite integrals using Fourier transform.
C201.5	Calculate the Z-transform of discrete time systems and solve the difference equations using Z-transform.
Course Code & Title : EE6301 Digital Logic Circuits Year/ Semester : II Year/ III Semester Course Index : C202	
C202.1	Solve the problems in number systems and can examine the important characteristics of the digital logic family circuits.
C202.2	Develop combinational logic circuits with minimal hardware complexity.
C202.3	Construct various synchronous and asynchronous circuits with its application.
C202.4	Apply the concepts of asynchronous sequential circuits and PLAs.
C202.5	Perform a software platform for numerous digital circuits using VHDL.
Course Code & Title : EE6302 Electromagnetic Theory Year/ Semester : II Year/ III Semester Course Index : C203	
C203.1	Apply vector calculus to static electric-magnetic fields in different engineering situations.
C203.2	Explain electric potential problems in linear, isotropic media for simple geometrics.
C203.3	Determine voltage gradients for simple charge and current configurations and the force between charges and currents.
C203.4	Analyze Maxwell's equation in different forms (differential and integral) and apply them to diverse engineering problems.
C203.5	Examine the phenomena of wave propagation in free space and different media and its interfaces.
Course Code & Title : GE6351 Environmental Science and Engineering Year/ Semester : II Year/ III Semester	

Course Index : C204	
C204.1	Understand the importance of public awareness on environment and nature of biodiversity.
C204.2	Know about the various causes, effect and control measures of environmental pollution.
C204.3	Comprehend the human development that leads to environmental disasters, the value of natural resources and their conservation.
C204.4	Recognize the value of public participation in environmental protection, Environmental Management and Legislation Acts and sustainable development.
C204.5	Learn the problems related to population and their remedial measures.
Course Code & Title : EC6202 Electronic Devices and Circuits Year/ Semester : II Year/ III Semester Course Index : C205	
C205.1	Explain the structure, operation and characteristics of the basic PN junction diode and special diodes.
C205.2	Determine the characteristics and biasing circuits for transistor and thyristor devices.
C205.3	Develop BJT and FET small signal amplifier models.
C205.4	Construct applications using the basic electronic devices.
C205.5	Apply the concept of feedback to improve the stability of circuits.
Course Code & Title : EE6303 Linear Integrated Circuits and Applications Year/ Semester : II Year/ III Semester Course Index : C206	
C206.1	Discuss the fabrication process of different IC's.
C206.2	Determine the DC and AC characteristics of op-amps.
C206.3	Construct linear and nonlinear applications of op-amps.
C206.4	Examine the operation of special function IC's.
C206.5	Explain the operation of application IC's like voltage regulator and switching regulator.
Course Code & Title : EC6361 Electronics Laboratory Year/ Semester : II Year/ III Semester Course Index : C207	
C207.1	Explain the fundamentals of operation and characteristics of semiconductor devices.
C207.2	Calculate the basic parameters of semiconductor devices and their limiting factors.
C207.3	Demonstrate BJT amplifiers in various configurations including DC biasing techniques.
C207.4	Determine the frequency response characteristics of amplifiers.
C207.5	Construct basic electronic circuits, particularly with application to diodes, field-effect transistors and bipolar junction transistors.
Course Code & Title : EE6311 Linear and Digital Integrated Circuits Laboratory Year/ Semester : II Year/ III Semester Course Index : C208	
C208.1	Solve boolean functions and develop adder, subtractor circuits.
C208.2	Use various code converters to understand the importance of code conversion.
C208.3	Examine the operation of 4 - bit shift registers.

C208.4	Acquire knowledge on application of Op-Amp in various applications.
C208.5	Construct counters using specific counter IC.
Course Code & Title : MA6459 Numerical Methods Year/ Semester : II Year/ IV Semester Course Index : C209	
C209.1	Solve the algebraic, transcendental and system of linear equations.
C209.2	Apply the Interpolation and approximations in various problems.
C209.3	Find the differentiations and integration using numerical tools.
C209.4	Determine the solution of initial value problems for ordinary differential equations.
C209.5	Determine the solution of boundary value problems in ordinary and partial differential equations
Course Code & Title : EE6401 Electrical Machines-I Year/ Semester : II Year/ IV Semester Course Index : C210	
C210.1	Apply techniques of magnetic-circuit analysis and introduce magnetic materials.
C210.2	Explore the operation and performance of single and three phase transformer.
C210.3	Demonstrate the concepts of electromechanical energy conversion and understand the concepts in rotating machines.
C210.4	Explain the working principles and performance of DC generator.
C210.5	Illustrate the working principles, starting, speed control and performance of DC motor.
Course Code & Title : CS6456 Object Oriented Programming Year/ Semester : II Year/ IV Semester Course Index : C211	
C211.1	Explain the difference between object oriented programming and procedural programming.
C211.2	Implement programs using C++ features such objects, operator overloading, dynamic memory allocation, and polymorphism.
C211.3	Develop applications using Object Oriented Programming concepts such as inheritance and Generic programming using C++.
C211.4	Implement simple Java programs using object-oriented concepts like classes, objects and inheritance.
C211.5	Apply exception handling and implement interfaces, packages, strings, I/O streams to various computing problems using Java.
Course Code & Title : EE6402 Transmission and Distribution Year/ Semester : II Year/ IV Semester Course Index : C212	
C212.1	Explain the basics of transmission and distribution in a power system and the operation of the different distribution schemes.
C212.2	Derive the expressions for the computation of transmission line parameters.
C212.3	Develop an equivalent circuit for the transmission lines based on distance and calculate the operating voltage for determining voltage regulation and efficiency.
C212.4	Calculate the voltage distribution in insulator strings and cables and methods to improve the same.
C212.5	Determine the mechanical design parameters of transmission lines.
Course Code & Title : EE6403 Discrete Time Systems and Signal Processing	

Year/ Semester : II Year/ IV Semester	
Course Index : C213	
C213.1	Analyze the properties of signals and systems.
C213.2	Analyze discrete time LTI systems using Z transform and DTFT.
C213.3	Apply the concepts of DFT and FFT to solve Electrical Engineering Problems.
C213.4	Design FIR and IIR digital filters.
C213.5	Describe the concepts of DSP Processors.
Course Code & Title : EE6404 Measurements and Instrumentation	
Year/ Semester : II Year/ IV Semester	
Course Index : C214	
C214.1	Determine various types of errors present in measurements.
C214.2	Explain the various types of electrical and electronics instruments.
C214.3	Compare a suitable measuring instrument used for measuring different electrical quantities.
C214.4	Explain the operating principles of different storage and display devices.
C214.5	Classify and select transducer for application in an electrical system.
Course Code & Title : CS6461 Object Oriented Programming Laboratory	
Year/ Semester : II Year/ IV Semester	
Course Index : C215	
C215.1	Implement the features of object oriented programming to solve real world problems.
C215.2	Design and implement the java programs for simple applications.
C215.3	Implement the Java programs with the help of interface and packages.
C215.4	Develop applications using object oriented programming concepts.
C215.5	Become familiar with the fundamentals and acquire programming skills in the Java language.
Course Code & Title : EE6411 Electrical Machines Laboratory-I	
Year/ Semester : II Year/ IV Semester	
Course Index : C216	
C216.1	Determine the performance of various D.C. Generators and understand the applications of it to power system
C216.2	Troubleshoot the operation of various D.C. Generators
C216.3	Draw the performance characteristics of various D.C. Motors and understand the applications of it to power system
C216.4	Predetermine the performance of DC motor by conducting suitable tests
C216.5	Calculate the voltage regulation and predetermine the performance of the single phase and three phase transformer
Course Code & Title : EE6501 Power System Analysis	
Year/ Semester : III Year/ V Semester	
Course Index : C301	
C301.1	Classify the power system components such as generator, transformer and transmission lines and explain the concept of deregulated market.
C301.2	Derive steady state power flow using iterative techniques for planning, operation and control of a power system and use bus admittance matrix for power flow.
C301.3	Determine the short circuit capacity of any power system under symmetrical fault using bus impedance matrix and Thevenin's theorem concepts to calculate

	fault current.
C301.4	Apply suitable protective scheme against unsymmetrical faults in power system and explain various sequence diagrams of a power system under unsymmetrical fault condition.
C301.5	Explain the power system under transient condition and derive transient behaviour of the power system using numerical methods.
Course Code & Title : EE6502 Microprocessors and Microcontrollers Year/ Semester : III Year/ V Semester Course Index : C302	
C302.1	Explain the function of 8085 microprocessor and memory organization.
C302.2	Write a program using 8085 microprocessor instructions.
C302.3	Elaborate the operation of 8051 microcontroller architecture.
C302.4	Describe the commonly used peripherals and interfacing.
C302.5	Develop an applications using 8051 microcontroller.
Course Code & Title : ME6701 Power Plant Engineering Year/ Semester : III Year/ V Semester Course Index : C303	
C303.1	Describe the various systems of coal based thermal power plant.
C303.2	Explain the various cycles and systems in diesel, gas and combined cycle power plant.
C303.3	Discuss the working of nuclear reactors in nuclear power plant.
C303.4	Explain the principle/construction and working of hydroelectric and various non-conventional power plants.
C303.5	Examine energy, economic and environmental issues of power plant.
Course Code & Title : EE6503 Power Electronics Year/ Semester : III Year/ V Semester Course Index : C304	
C304.1	Explain the basic operation and the characteristics of various Power Electronics devices like SCR, TRIAC, DIAC, IGBT, GTO etc.
C304.2	Apply the knowledge of aforesaid Power semiconductor devices for designing various applications related to rectifiers.
C304.3	Classify the chopper circuits and explain their applications in various fields.
C304.4	Understand and analyze various inverter circuits and PWM techniques.
C304.5	Explore the single phase and three AC voltage-controllers and their applications.
Course Code & Title : EE6504 Electrical Machines -II Year/ Semester : III Year/ V Semester Course Index : C305	
C305.1	Explain the concept, principle and performance of salient and non-salient pole synchronous generator.
C305.2	Describe the concept, principle and performance of synchronous motor.
C305.3	Illustrate the construction, working principle and performance of three phase induction motor.
C305.4	Examine various types of starting and speed control methods of three phase induction motor.
C305.5	Analyze the principle and performance of single phase induction motors and special electrical machines.
Course Code & Title : IC6501 Control systems	

Year/ Semester : III Year/ V Semester	
Course Index : C306	
C306.1	Develop transfer function models of electrical systems and mechanical systems.
C306.2	Derive the time response of systems and analyse the behaviour of the system.
C306.3	Draw the frequency response of the system and describe the effect of compensation on frequency response.
C306.4	Apply the concepts of system stability to analyse performance of closed loop systems.
C306.5	Explain the basic concepts of state variable analysis of systems and evaluate the solution of state and output equation.
Course Code & Title : EE6511 Control and Instrumentation Laboratory	
Year/ Semester : III Year/ V Semester	
Course Index : C307	
C307.1	Apply basic Science, circuit theory, Electro-magnetic field theory, control theory concepts for engineering problems
C307.2	Acquire the basic knowledge about the measurements and compare the characteristics of measuring devices.
C307.3	Explore the working of signal conditioning circuits.
C307.4	Develop mathematical models of electrical systems.
C307.5	Choose controller parameters based on tuning methods.
Course Code & Title : EE6512 Electrical Machines Laboratory-II	
Year/ Semester : III Year/ V Semester	
Course Index : C308	
C308.1	Determine the regulation of a synchronous generator and explain the applications to power system.
C308.2	Draw the performance characteristics of synchronous motor and explain the applications to power system.
C308.3	Draw the performance characteristics of three phase squirrel cage and slip ring induction motor and understand the applications to power system.
C308.4	Calculate the efficiency and performance characteristics of single phase induction motor.
C308.5	Troubleshoot the operation of various Induction and synchronous machine.
Course Code & Title : GE 6674 Communication and Soft Skills – Laboratory	
Year/ Semester : III Year/ V Semester	
Course Index : C309	
C309.1	Identify and interpret visuals, communicate in formal and informal conversations, give presentations, and participate in GD.
C309.2	Explain reading comprehension passages of higher levels, draft Resume, cover letter, reports, emails, and write blogs.
C309.3	Differentiate between IELTS & TOEFL and take placement oriented verbal ability tests.
C309.4	Demonstrate appropriate verbal, non-verbal and paralinguistic skills while attending job interviews.
C309.5	Show leadership traits, team skills and essential soft skills and efficiency to excel as a professional.
Course Code & Title : EC6651 Communication Engineering	
Year/ Semester : III Year/ VI Semester	

Course Index : C310	
C310.1	Apply various analog communication techniques in all communication systems.
C310.2	Apply various digital communication techniques in all communication systems.
C310.3	Apply Source and Error control coding in both wired and wireless communication systems.
C310.4	Analyze the various multiple access techniques.
C310.5	Explain the basic concepts used in satellite and optical fiber communications.
Course Code & Title : EE6601 Solid State Drives Year/ Semester : III Year/ VI Semester Course Index : C311	
C311.1	Explain the basics of dynamics and steady state stability of electric drive.
C311.2	Describe the operation of converter/chopper fed dc drive.
C311.3	Apply the control methods for the induction motor drives.
C311.4	Analyze the performance of synchronous motor drives.
C311.5	Design and apply the current and speed controllers for drives.
Course Code & Title : EE6602 Embedded Systems Year/ Semester : III Year/ VI Semester Course Index : C312	
C312.1	Explain the building blocks of embedded systems
C312.2	Analyse the various communications in processors and input/output interfacing
C312.3	Explain the embedded development strategies
C312.4	Apply the various scheduling techniques in Real Time Operating System
C312.5	Apply the concepts of embedded system in application developments within realistic constraints such as economic, environmental, social and ethical
Course Code & Title : EE6603 Power System operation and Control Year/ Semester : III Year/ VI Semester Course Index : C313	
C313.1	Demonstrate the basic concepts of power system operation and control and apply the concept of load forecasting to predict the future energy requirement.
C313.2	Construct the power-frequency controller to single area and two area systems.
C313.3	Employ the suitable control actions to maintain the voltage profile by controlling reactive power against various types of load.
C313.4	Schedule the generators in power system economically by means of unit commitment and economic dispatch solutions.
C313.5	Apply the concept of computer control of power system to improve the control of power system.
Course Code & Title : EE6604 Design of Electrical Machines Year/ Semester : III Year/ VI Semester Course Index : C314	
C314.1	Explain about basic design concepts and various rating of electrical machines.
C314.2	Describe the design of various components of D.C. electrical machines.
C314.3	Discuss the design of overall dimensions of the transformer.
C314.4	Design the stator and types of rotor of three phase induction motor.
C314.5	Analyze the constraints and design the vital parts of Synchronous machines.
Course Code & Title : EE6002 Power System Transients	

Year/ Semester : III Year/ VI Semester	
Course Index : C315	
C315.1	Explore the source and effects of power system transients
C315.2	Explain the generation of switching transients and effects
C315.3	Explore the mechanism of lightning transients and lightning protection methods
C315.4	Analyze the travelling wave concepts
C315.5	Apply the travelling waves in integrated system
Course Code & Title : EE6611 Power Electronics and Drives Laboratory	
Year/ Semester : III Year/ VI Semester	
Course Index : C316	
C316.1	Demonstrate the gate pulse generation using various configurations.
C316.2	Explain the characteristics of power electronic switching devices.
C316.3	Compare the different power converters and its applications.
C316.4	Interpret the performance of AC voltage controller.
C316.5	Simulate various power electronic converter circuits.
Course Code & Title : EE6612 Microprocessors and Microcontrollers Laboratory	
Year/ Semester : III Year/ VI Semester	
Course Index : C317	
C317.1	Apply the arithmetic operations, logic operations and sorting using 8085 microprocessor.
C317.2	Write a program for code conversion, finding the maximum and minimum numbers in a series.
C317.3	Analyze the operations of peripheral interfacing with 8085 microprocessor.
C317.4	Apply the arithmetic, logic operations and branching operation using 8051 microcontroller.
C317.5	Write a program for peripheral interfacing with 8051 microcontroller.
Course Code & Title : EE6613 Presentation Skills and Technical Seminar	
Year/ Semester : III Year/ VI Semester	
Course Index : C318	
C318.1	Understand the significance of learning new topics in electrical engineering discipline.
C318.2	Prepare and present the report on technical advancements and developments in electrical engineering.
C318.3	Organize the presentation using the concepts of ordering and determining the central, main and supporting ideas.
C318.4	Communicate effectively and clearly in front of peers and faculty members.
C318.5	Perform well in the placement interviews with good communication skills.
Course Code & Title : EE6701 High Voltage Engineering	
Year/ Semester : IV Year/ VII Semester	
Course Index : C401	
C401.1	Investigate the source, effects and protection methods of over voltages in power system
C401.2	Explore the breakdown mechanism in different dielectrics
C401.3	Explain the different methods of overvoltage generation
C401.4	Explore the various overvoltage measurement methods
C401.5	Apply the concepts of high voltage power apparatus testing procedures
Course Code & Title : EE6702 Protection and Switchgear	

Year/ Semester : IV Year/ VII Semester	
Course Index : C402	
C402.1	Analyze the knowledge on different protective schemes in power system.
C402.2	Explain various electromagnetic relays and its application.
C402.3	Classify the protection scheme for various faults in motor, generator, transformer, bus bar, transmission line.
C402.4	Examine various static relays and numerical relays and its application.
C402.5	Describe concepts and principle of various circuit breakers.
Course Code & Title : EE6703 Special Electrical Machines	
Year/ Semester : IV Year/ VII Semester	
Course Index : C403	
C403.1	Explain the variable reluctance principle, different types of rotor construction and performance of synchronous reluctance motors.
C403.2	Describe the different types of stepper motors, closed loop control and its applications.
C403.3	Illustrate the sensor less operation and performance of switched reluctance motors using mathematical model.
C403.4	Classify the various types of permanent magnet materials used for Brushless D.C. motors and explain its performance.
C403.5	Explain the working of ideal PMSM, its performance along with power controllers used for various applications.
Course Code & Title : MG6851 Principles of Management	
Year/ Semester : IV Year/ VII Semester	
Course Index : C404	
C404.1	Explain the evolution of management and the functions and roles of managers.
C404.2	Exercise the different types of planning process and tools used for planning
C404.3	Apply the concept of organization structures and the functions of human resource manager in a management system.
C404.4	Classify the different theories of motivation and leadership.
C404.5	Illustrate the fundamental elements of controlling
Course Code & Title : EE6004 Flexible AC Transmission Systems	
Year/ Semester : IV Year/ VII Semester	
Course Index : C405	
C405.1	Calculate the rating of the compensators using the concept of passive and active reactive power control techniques.
C405.2	Employ different Static VAR compensators for various applications
C405.3	Illustrate various modes in thyristor controlled series capacitors and determine the rating of compensators.
C405.4	Solve the reactive power problem by active compensators and design the suitable scheme for different applications
C405.5	Apply different techniques to co-ordinate multiple FACTS controllers and study the interaction between FACTS controllers.
Course Code & Title : EE6005 Power Quality	
Year/ Semester : IV Year/ VII Semester	
Course Index : C406	
C406.1	Illustrate the power quality problems.
C406.2	Explore the sources for sags in power system.

C406.3	Classify the power quality problem due to over voltages.
C406.4	Explain the various sources for harmonics and harmonics reduction methods.
C406.5	Apply the expert system for power quality monitoring.
Course Code & Title : EE6008 Microcontroller Based System Design Year/ Semester : IV Year/ VII Semester Course Index : C407	
C407.1	Explain the functions of the blocks in PIC microcontroller.
C407.2	Apply the different Interrupts and Timers in PIC microcontroller.
C407.3	Describe the concepts of Peripheral interfacing with PIC microcontroller.
C407.4	Explain the programmer's model of ARM processor.
C407.5	Analyze the applications of ARM processor.
Course Code & Title : EE6711 Power System Simulation Laboratory Year/ Semester : IV Year/ VII Semester Course Index : C408	
C408.1	Develop a suitable program for determining the transmission line parameters and its performance evaluation.
C408.2	Simulate the power system to obtain the power flow solution using numerical iterative methods.
C408.3	Perform fault analysis for a given power system under symmetrical and unsymmetrical fault.
C408.4	Analyze the stability of the power system by simulation using appropriate techniques
C408.5	Develop a suitable program to solve economic dispatch problem and load frequency dynamics of interconnected power system.
Course Code & Title : EE6712 Comprehension Year/ Semester : IV Year/ VII Semester Course Index : C409	
C409.1	Describe the basic concepts of electrical and electronics subjects.
C409.2	Solve objective type questions in the field of electrical and electronics engineering
C409.3	Review, prepare and present technological developments
C409.4	Analyse the modern trends in the field of electrical and electronics engineering.
C409.5	Answer effectively during technical interviews.
Course Code & Title : EE6801 Electric Energy Generation, Utilization and Conservation Year/ Semester : IV Year/ VIII Semester Course Index : C410	
C410.1	Illustrate the various concepts of electric drives and the mechanism of train movement in electric traction system.
C410.2	Apply the energy efficient concept in illumination and lighting schemes.
C410.3	Describe the concepts of electric heating and electric welding.
C410.4	Explain the principle of Solar Radiation and Solar Energy Collectors.
C410.5	Analyze the principles and utilization of wind energy.
Course Code & Title : EE6009 Power Electronics for Renewable Energy Systems Year/ Semester : IV Year/ VIII Semester	

Course Index : C411	
C411.1	Describe the environmental aspects of electrical energy generation and utilization.
C411.2	Classify the various electrical machines for renewable energy applications.
C411.3	Explain the different power converters namely DC to AC, AC to DC, DC to DC and AC to AC converters for renewable energy sources.
C411.4	Compare and analyze the various operating modes of wind electrical generators and solar energy systems.
C411.5	Analyze the different type of Hybrid renewable energy system.
Course Code & Title : GE6075 Professional Ethics in Engineering Year/ Semester : IV Year/ VIII Semester Course Index : C412	
C412.1	Apply ethics, morals and human values in society.
C412.2	Explain about engineering ethics.
C412.3	Realize the responsibilities of engineers as experimenters.
C412.4	Recognize the safety, risks, risk benefit analysis and rights of an engineer.
C412.5	Discuss the importance of the global issues, moral leadership and code of conduct.
Course Code & Title : GE6757 Total Quality Management Year/ Semester : IV Year/ VIII Semester Course Index : C413	
C413.1	Discuss the contributions of Quality Guru
C413.2	Explain the principles of TQM
C413.3	Apply the tools and techniques of quality management to manufacturing and service processes
C413.4	Describe TQM tools and techniques such as Control Charts, QFD and TPM.
C413.5	Discuss the elements of Quality system standards
Course Code & Title : EE6811 Project work Year/ Semester : IV Year/ VIII Semester Course Index : C414	
C414.1	The students will be able to apply the fundamental knowledge and skills, which are acquired within the technical area, to a given problem
C414.2	- The students will be able to identify and summarize an appropriate list of literature review, analyze previous researchers' work and relate them to the project - Within given constraints, even with limited information, the students will be able to independently analyze and discuss complex inquiries/problems and handle larger problems on the advanced level within the technical area.
C414.3	- The students will be able to design engineering solutions to complex problems in a systematic approach. - The students will be able to identify and apply appropriate parameters, assumptions and design criteria in consideration of health and safety (example: the use of codes of practice), ethics, economics, environment, sustainability
C414.4	The students will be able to apply research and conduct experiments, as well as to analyze and interpret data that yield the results and answer important

	applicable research questions.
C414.5	The students will be able to utilize technology tools for communication, collaboration, information management, and decision support.
C414.6	The students will be able to demonstrate the knowledge, skills and attitudes of a professional engineer.
C414.7	The students will be able to interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment.
C414.8	The students will be able to demonstrate a strong working knowledge of ethics and professional responsibility.
C414.9	The students will be able to document and present one's own work, for a given target group, with strict requirements on structure, format, and language usage.
C414.10	The students will be able to present the project outlining the approach and expected results using good oral and written presentation skills.
C414.11	The students will be able to demonstrate effective organizational leadership and change skills for managing projects and project teams.
C414.12	The students will be able recognize the need for life-long learning by undergoing the project work.